How the Metrics Are Computed

AI Use Impact Tracker — Technical Reference

Sapien Labs | May 2026

This document is a detailed walk-through of every metric the AI Impact Tracker publishes, including the exact formulas, denominators, suppression rules, and how the dashboard's Period filter pools months together.

It is a companion to **COLUMNS.md** (which documents the source-side survey columns) and **TRACKER.md** (the spec/contract for the metric layer). All formulas match the implementation in `metrics.py` and `normalize.py`.

1. Building Blocks

Every metric is computed from one row per respondent per month. Three derived fields drive almost everything else.

ai_freq_int — AI-use frequency (0–6 ordinal scale)

Mapped from the raw ai_freq text answer:

Integer	Label	Raw text mapped
0	Never	"I have never used an AI assistant"
1	Rarely	"Rarely"
2	Monthly	"A few times a month"
3	Weekly	"Several days a week"
4	Daily	"Several times a day"
5	Constantly	"Constantly", "Constamment", "Constantly (when awake)"
6	Always	"All of the time"
—	null	"N/A" or anything not recognised

is_user — does this respondent use AI at all?

$$is_{user} = \mathbb{1}¿$$

A respondent at level 0 (“Never”) is not an AI user.

Impact flags — one-hot from ai_impact_work

The pipe-delimited ai_impact_work answer is split into nine boolean columns; each respondent can have multiple flags set true.

Flag	Triggered by answer
impact_improved_quality	“Improved my work quality or output”
impact_new_opportunities	“Created new job or income opportunities”
impact_adaptation_pressure	“Increased pressure to adapt or work faster”
impact_job_anxiety	“Made me worry about the future of my job or industry”
impact_job_loss	“Caused me to lose my job”
impact_reduced_income	“Reduced my income or made it harder to find work”
impact_none	“No impact”
impact_other	“Another impact not listed here” (plus any unknown raw option)
impact_not_sure	“Not sure”
impact_na	Raw value was null / N/A / blank

Two derived boolean columns roll the flags into a positive / negative direction:

$$is_{positive} = impact\_improved_{quality} \vee impact\_new_{opportunities}$$

$$is_{negative} = impact\_adaptation_{pressure} \vee impact\_job_{anxiety} \vee impact\_job_{loss} \vee impact\_reduced_{income}$$

impact_none, impact_other, impact_not_sure, and impact_na are treated as neither positive nor negative.

2. Core Indicators

These are reported for every stratum × month.

n_respondents

Plain count of respondents in the cell:

$$N = |\{ i : i \in stratum \times month \}|$$

This is also the denominator for adoption_rate and freq_mean.

adoption_rate — share who use AI at all

$$adoption_{rate} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[is\_user_i]$$

Range [0, 1]. Includes respondents at every ai_freq level from “Rarely” upward.

freq_mean — mean AI-use frequency (0–6 scale)

$$freq_{mean} = \frac{1}{N'} \sum ai\_freq\_int_i$$

Range [0, 6]. A freq_mean of 2.08 means roughly “monthly use” on average; 4.0 means “daily” on average.

Note: *freq_mean is computed and stored, but no longer surfaced as a selectable map metric on the redesigned dashboard.*

freq_share$_0$ … freq_share$_6$ — distribution

For each ordinal level k = 0, …, 6:

$$freq\_share_k = \frac{1}{N'} \sum \mathbb{1}[ai\_freq\_int_i = k]$$

The seven shares sum to 1.

3. The Impact Denominator

The impact metrics in §4–§6 do not use n_respondents as their denominator. They use a stricter base population, the impact denominator:

$$\text{¿}\_impact\_denom_i = is\_user_i \wedge has\_impact\_response_i$$

where has_impact_response means at least one of the eight real impact flags is true (all flags except impact_na):

$$has\_impact\_response = \text{¿} f \in \mathcal{F} \text{¿}\{impact\_na\}\text{¿❑} flag_e$$

i.e. the disjunction over impact_improved_quality, impact_new_opportunities, impact_adaptation_pressure, impact_job_anxiety, impact_job_loss, impact_reduced_income, impact_none, impact_other, and impact_not_sure.

So the impact denominator is reported as `n_impact_denominator` and equals the count of respondents who:

1. use AI at all (ai_freq_int ≥ 1), and
2. answered the impact question (gave at least one option, including “No impact” or “Not sure”).

Respondents who never use AI, or whose impact answer was blank/N/A, are excluded. The dashboard tooltip describes this base as “AI users with impact response.”

4. Impact Share Metrics

For each impact flag listed in §1, the share is computed against the impact denominator. Let N_4 denote n_impact_denominator:

$$impact\_share_f = \frac{1}{N_4} \sum \mathbb{1}[flag(f, i)]$$

The metrics published are:

Metric	Numerator flag
impact_share_improved_quality	impact_improved_quality
impact_share_new_opportunities	impact_new_opportunities
impact_share_adaptation_pressure	impact_adaptation_pressure
impact_share_job_anxiety	impact_job_anxiety
impact_share_job_loss	impact_job_loss
impact_share_reduced_income	impact_reduced_income
impact_share_none	impact_none
impact_share_other	impact_other
impact_share_not_sure	impact_not_sure

Each is a value in [0, 1].

The shares do not have to sum to 1. Respondents can select more than one impact, so a given respondent can contribute to multiple shares. A respondent who picked “Improved my work quality” and “Created new opportunities” counts toward both impact_share_improved_quality and impact_share_new_opportunities.

5. Positive / Negative / Net Impact

Two rolled-up shares are computed on the same impact denominator:

$$positive\_impact_{share} = \frac{1}{N_4} \sum \mathbb{1}[is\_positive_i]$$

$$negative\_impact_{share} = \frac{1}{N_4} \sum \mathbb{1}[is\_negative_i]$$

where is_positive and is_negative are defined in §1.

Net Impact Index (NII) — the difference:

$$NII = positive\_impact_{share} - negative\_impact_{share}$$

Range [−1, +1]:

- +1 means every respondent in the cell picked only positive impacts and no negative ones.
- −1 means every respondent picked only negative impacts.
- 0 means equal positive and negative shares (by cancellation or because everyone answered "No impact" / "Not sure").

A respondent who picks one positive and one negative contributes +1 to both shares, so the net for that respondent's contribution is 0. The index is a population-level summary, not a per-respondent score.

Note: *net_impact_index is computed and stored, but no longer surfaced as a selectable map metric on the redesigned dashboard. It still drives the dose-response panel (§7).*

6. Weighted Impact Index

The **Weighted Impact Index** (weighted_impact_index) is the dashboard’s headline metric. Unlike the simple net index in §5, it assigns each impact flag a signed weight reflecting how positive or negative that outcome is, then averages a per-respondent score across the impact denominator.

Weights

These are editorial values set by the Sapien Labs team; they reflect judgment about relative severity, not an empirical calibration.

Flag	Weight
impact_new_opportunities	**+1.00**
impact_improved_quality	**+0.50**
impact_job_anxiety	**−0.25**
impact_adaptation_pressure	**−0.50**
impact_reduced_income	**−0.75**
impact_job_loss	**−1.00**
impact_none, impact_other, impact_not_sure	0

Per-respondent score

For each respondent i in the impact denominator, sum the weights of every flag they selected:

$$s_i = \sum_{f \in \mathcal{F}} w_e \cdot \mathbb{1}[flag(f, i)]$$

where $w_e = \text{IMPACT\_WEIGHTS}[f]$ and $\mathcal{F}$ is the set of eight weighted flags (excluding impact_na).

Examples:

- A respondent who selected only “Improved my work quality” scores +0.5.
- A respondent who selected “Improved my work quality” and “Adaptation pressure” scores +0.5 + (−0.5) = 0.
- A respondent who selected “Job loss” and “Reduced income” scores −1.0 + (−0.75) = −1.75.
- A respondent who selected “No impact” scores 0.

A single respondent’s score can exceed ±1 if they selected several weighted flags in the same direction.

Cell value

The published weighted_impact_index is the arithmetic mean of these per-respondent scores over the impact denominator:

$$WII = \frac{1}{N_4} \cdot \sum s_i$$

Range is roughly [−1, +1] in practice but is not symmetric: a respondent would need to select multiple negative flags simultaneously to drive a cell well below −1.

A positive cell value means the average AI user in that stratum had a net-positive experience; a negative value means the average user had a net-negative experience.

7. Dose-Response

The dashboard’s “Dose-response” row shows the Net Impact Index by AI-use frequency level. For each stratum × month, and for each ordinal ai_freq level k = 1, …, 6:

$$dose\_response[k] = pos\_share_k - \neg\_share_k$$

The denominator is respondents in the impact denominator (§3) at that specific frequency level. If that level has fewer than MIN_N respondents (§9), the entry is null (rendered as n/a in the dashboard).

Level 0 (“Never”) is not part of the impact denominator and is not reported.

Interpretation: a dose-response that climbs from +0.10 at “Rarely” to +0.40 at “Daily” suggests the net impact is more positive among heavier users — a hint of dose-dependence, not a causal claim.

8. Confidence Intervals

Share metrics — Wilson 95% score interval

All share metrics (`adoption_rate`, every `impact_share_*`, `positive_impact_share`, `negative_impact_share`) get a Wilson score 95% CI. Given k successes out of n trials and z = 1.96:

$$\hat{p} = \frac{k}{n}$$

$$\tilde{p} = \frac{\hat{p} + z^2/2n}{1 + z^2/n}$$

$$CI = \tilde{p} \pm \frac{z \cdot \sqrt{[\ \hat{p}(1-\hat{p})/n + z^2/4n^2\ ]}}{1 + z^2/n}$$

$$ci_{low} = max(0,\ CI\square_{low}), ci_{high} = min(1,\ CI\square_{high})$$

Wilson is preferred over the normal-approximation Wald interval because it stays inside [0, 1] even at extreme proportions and small n.

freq_mean — Wald 95% CI

$$SE = \frac{\sigma}{\sqrt{N'}}(ddof = 1)$$

$$CI = freq_{mean} \pm 1.96 \times SE$$

weighted_impact_index — Wald 95% CI

The per-respondent score s_i (§6) is treated as a continuous variable:

$$SE = \frac{\sigma}{\sqrt{N_4}} \left(ddof = 1 \right)$$

$$CI = WII \pm 1.96 \times SE$$

where σ_s is the sample standard deviation of the per-respondent scores and N_4 is the impact denominator size. *All CIs assume simple random sampling — they will be revised once survey weights are introduced.*

9. Minimum-N Suppression

Any stratum × month cell with fewer than **MIN_N (default 50)** respondents is flagged suppressed = true and its metric values are written as null. The dashboard does not display suppressed cells.

The threshold applies at the finest stratum level. So if a country × gender × age cell has 41 respondents, that cell is suppressed, even if the corresponding country × gender (without age) cell has 600.

MIN_N is configurable per run via the --min-n flag to main.py. The active value is stored in _meta.json and embedded in the rendered HTML as the suppression threshold shown in the dashboard subtitle.

The dashboard also applies a separate UI threshold: the Gender and Age band dropdowns only list categories that have at least one non-suppressed cell — so options with no usable data anywhere are hidden.

10. Stratification (Country × Gender × Age)

Every metric is computed independently for each of these eight stratum levels:

Level	Group-by columns (+ year, month)
global	—
country	country_clean
gender	gender_clean
age_band	age_band
country_gender	country_clean, gender_clean
country_age_band	country_clean, age_band
gender_age_band	gender_clean, age_band
country_gender_age_band	country_clean, gender_clean, age_band

The dashboard picks the finest level needed for the current filter combination so every displayed number is a precomputed cell value, not a JS-side aggregation:

Filters	Level used
no demographic filter	country

gender only	country_gender
age only	country_age_band
gender + age	country_gender_age_band

This is also why removing a demographic filter doesn't just average the cells back together — the dashboard re-loads the coarser stratum.

11. Rolling-Window Pooling (Period Filter)

The dashboard's Period selector — Single month / Last 3 / Last 6 / Last 12 — pools the precomputed monthly cells into a rolling window. Pooling is done in the browser on the embedded JSON and does not change the published Parquet output.

For each (country × gender × age) cell across the window of months, fields are pooled with the following rules:

Field	Aggregation	Weight
n_respondents, n_impact_denominator	sum	—
adoption_rate, freq_mean	weighted mean	n_respondents
weighted_impact_index, net_impact_index, all impact_share_*, positive/negative_impact_share	weighted mean	n_impact_denominator
dose_response[k] (k = 1…6)	weighted mean	n_impact_denominator

This is approximate. The exact pooled value would require re-running the metric layer on the pooled respondent-level data. The dashboard's weighted-mean of monthly aggregates is correct in expectation but ignores within-month variance and does not produce new confidence intervals.

Cells suppressed in a given month (originally below MIN_N) are dropped before pooling, so a country can appear in the rolling window even if some of its constituent months were individually suppressed.

12. Summary Card (Dashboard)

The top-right summary card on the dashboard is a fixed reference and is not affected by Month, Period, Country, Gender, or Age filters. It always shows:

3. The global weighted average of the currently-selected metric over the trailing 3 full months.
4. The total respondents over those 3 months.
5. The total respondents across the entire dataset to date.

“Full” means non-partial. The dashboard detects the latest partial month by comparing each month’s total respondents against the median of prior months — any month with respondents below 60% × median is considered still in-flight and excluded from the trailing-3 window. This is the same rule used to mark the partial point in the trend chart.

Only the metric dropdown updates this card; switching metric swaps the displayed label and value but does not change the underlying 3-month window.

13. Known Limitations

These limitations apply to every metric above.

13.1 No survey weights (non-probability sample)

The GMP respondent pool is a convenience sample recruited via online ads (Google Display, Meta, and organic search). It carries no post-stratification or design weights. Demographic proportions in the data do not match national population proportions — countries, age groups, and genders are represented according to who encounters and completes the survey, not according to census benchmarks.

All metrics are descriptive statistics of the responding population, not population-level estimates. Associations between ai_freq and impact outcomes are correlational; higher-frequency AI users who report better outcomes may differ systematically from lower-frequency users in education, occupation, digital literacy, or other unmeasured confounders. No causal claims are supported.

> *Planned mitigation: Demographic-weighted averaging using UN World Population Prospects age–sex distributions is on the roadmap. This would rebalance age and gender composition within each country but would not address self-selection or unmeasured confounders.*

13.2 Composition effects in time-series

Month-over-month changes may reflect shifts in who is responding rather than genuine attitudinal change. If a recruitment campaign increases responses from a younger demographic in one month, and younger respondents tend to report more positive AI impacts, the global WII will rise even if no individual changed their view. Country mix changes and seasonal patterns in survey completion can similarly shift metrics.

The stratified breakdowns (by country, gender, age) partially mitigate this by holding one dimension constant, but cross-dimensional composition effects remain. Rolling-window views help smooth short-term fluctuations.

13.3 Weight selection

The nine IMPACT_WEIGHTS (§6) were chosen by stakeholder judgment, not derived from empirical data. Different weight schemes would produce different index values and could alter relative country rankings. Sensitivity analysis (varying weights within a plausible range) has not yet been conducted. The asymmetric design — total negative weight (−2.5) exceeding total positive weight (+1.5) — is intentional but should be disclosed when comparing WII values across studies.

13.4 Rolling-window approximation

Rolling-window pooling (§11) uses total n_respondents as approximate weights. The exact calculation would re-aggregate from individual-level data within the pooled window. The approximation introduces minor error (±1 percentage point within 3–6 month windows) because the impact denominator differs slightly from total N. This trade-off is accepted for client-side performance.

13.5 Dose-response uses NII, not WII

The dose-response curves (§7) show the Net Impact Index by frequency level, not the Weighted Impact Index. Adding weighted dose-response is a planned enhancement.

13.6 Data source filtering not yet implemented

Per Tara’s specification, the production version should restrict the respondent pool to Google Display and Meta traffic sources and down-weight organic/search traffic to 10%. This is not yet implemented; the current pipeline processes all traffic sources equally.

13.7 Multi-select interaction effects

The WII treats each flag independently — a respondent selecting both “improved quality” (+0.5) and “job anxiety” (−0.25) receives the sum (+0.25). This additive model does not capture potential interactions between simultaneous positive and negative experiences.

13.8 Suppression and small-cell noise

Strata with fewer than 50 respondents (MIN_N) are suppressed entirely. For strata just above the threshold (e.g., N = 55), confidence intervals are wide and point estimates can shift substantially with small data changes.

13.9 CIs assume simple random sampling

Confidence intervals will be revised once survey weights are introduced.